# Supplementary Material for "When Are Triangles Invisible? Fundamental Limits of Higher-Order Structure Identifiability from Edge Flows"

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This supplement extends the main paper in three directions: **S1** Fano-type converses for *joint* support recovery (including a signal-distribution-agnostic bound and a sub-Gaussian achievability counterpart); **S2** identifiability under *partial edge observation*; **S3** *plug-in* estimation of the variance parameters and a fully adaptive detector. All statements are validated numerically by the released test-suite; Section S4 maps every figure to its script. Notation follows the main paper: $N$ snapshots, candidate set $\mathcal{T}$ with $p = |\mathcal{T}|$, latent support $\mathcal{S}$ with $|\mathcal{S}| = k$, curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$, per-coordinate null/alternative variances $v_0 = 3\sigma_n^2$, $v_1 = v_0(1+\rho)$. Throughout S1–S2 the candidates are pairwise edge-disjoint, so the curl coordinates are independent given $\mathcal{S}$ and the triangle Gram matrix is $\boldsymbol{G} = 3\boldsymbol{I}$.

## S1 Fano-type converses for joint support recovery

The main paper's Theorem 1 is a *single-triangle* converse: it bounds the error of deciding one triangle's activity and yields the floor $\rho^\star(N) = \Theta(\sqrt{\log(1/\delta)/N})$. Recovering the whole support $\mathcal{S}$ among $\binom{p}{k}$ possibilities is harder. The next two theorems quantify by how much.

We first record a reduction used by both proofs.

**Lemma S1.1** (Sufficiency of the curl coordinates)**.** *Consider the model of the main paper (Eq. (2)) with isotropic Gaussian noise $\boldsymbol{n}_t \sim \mathcal{N}(0, \sigma_n^2\boldsymbol{I})$, mutually independent $(\boldsymbol{a}_t, \boldsymbol{y}_t, \boldsymbol{h}_t, \boldsymbol{n}_t)$, and* arbitrary *laws for $(\boldsymbol{a}_t, \boldsymbol{y}_t, \boldsymbol{h}_t)$ (Gaussianity is needed only for the noise). Then*

$$I(\mathcal{S}; \{\boldsymbol{f}_t\}_{t=1}^N) \;=\; I(\mathcal{S}; \{\boldsymbol{c}_t\}_{t=1}^N), \qquad \boldsymbol{c}_t = \boldsymbol{B}_2^\top \boldsymbol{f}_t.$$

*Proof.* Decompose $\boldsymbol{f}_t$ orthogonally into its gradient, curl, and harmonic components. Because the isotropic Gaussian noise has independent projections onto orthogonal subspaces, and the gradient (resp. harmonic, curl) component of $\boldsymbol{f}_t$ equals $\boldsymbol{B}_1^\top \boldsymbol{a}_t$ (resp. $\boldsymbol{h}_t$, $\boldsymbol{B}_{2,\mathcal{S}}\boldsymbol{y}_t$) plus the corresponding independent noise projection, the three components are mutually independent, and only the curl component depends on $\mathcal{S}$. Hence the gradient and harmonic components are independent of $(\mathcal{S}, \text{curl component})$ and carry zero information about $\mathcal{S}$. Finally, on edge-disjoint candidates $\boldsymbol{B}_2$ has full column rank, so $\boldsymbol{c}_t$ is a bijective linear image of the curl component. $\square$

**Theorem S1.2** (Gaussian Fano converse)**.** *Let $\mathcal{S}$ be uniform on the $\binom{p}{k}$ supports of size $k$, with Gaussian signals and noise and edge-disjoint candidates. If an estimator achieves $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \le \varepsilon$, then*

$$N \;\ge\; \frac{(1-\varepsilon)\log\binom{p}{k} - \log 2}{k\,\mathrm{KL}_1(\rho)}, \qquad \mathrm{KL}_1(\rho) = \tfrac{1}{2}\big(\rho - \log(1+\rho)\big) \;\le\; \tfrac{\rho^2}{4}. \tag{1}$$

*In particular $\rho^2 N \gtrsim 4(1-\varepsilon)\log(p/k)$: joint recovery pays a $\log\binom{p}{k}$ factor where the single-triangle test pays $\log(1/\delta)$.*

*Proof.* By Lemma S1.1 and Fano's inequality, $\Pr[\hat{\mathcal{S}} \neq \mathcal{S}] \geq 1 - (I(\mathcal{S}; \boldsymbol{C}^N) + \log 2)/\log\binom{p}{k}$ with $\boldsymbol{C}^N = \{\boldsymbol{c}_t\}$. Bound the mutual information against the all-inactive reference law $P_0$: $I(\mathcal{S}; \boldsymbol{C}^N) \leq \frac{1}{\binom{p}{k}} \sum_{\mathcal{S}} \mathrm{KL}(P_{\mathcal{S}}^{\otimes N} \| P_0^{\otimes N}) = N\,\mathrm{KL}(P_{\mathcal{S}} \| P_0)$ for any fixed $\mathcal{S}$ by symmetry. Given $\mathcal{S}$, the coordinates of $\boldsymbol{c}$ are independent, equal in law to $\mathcal{N}(0, v_1)$ on $\mathcal{S}$ and $\mathcal{N}(0, v_0)$ off $\mathcal{S}$, and the reference has all coordinates $\mathcal{N}(0, v_0)$; hence $\mathrm{KL}(P_{\mathcal{S}} \| P_0) = k\,\mathrm{KL}(\mathcal{N}(0, v_1) \| \mathcal{N}(0, v_0)) = \frac{k}{2}\big(\frac{v_1}{v_0} - 1 - \log\frac{v_1}{v_0}\big) = k\,\mathrm{KL}_1(\rho)$. Rearranging Fano gives (1); the inequality $\mathrm{KL}_1 \leq \rho^2/4$ is $\log(1+\rho) \geq \rho - \rho^2/2$. □

**Theorem S1.3** (Signal-agnostic Fano converse)**.** *Keep the noise Gaussian, but let the active triangle signals $y_{\tau,t}$ be i.i.d. from an* arbitrary *zero-mean law with variance $\sigma_c^2$. Then any estimator with error at most $\varepsilon$ requires*

$$N \;\geq\; \frac{(1-\varepsilon)\log\binom{p}{k} - \log 2}{\frac{p}{2}\,\log\big(1 + \rho k/p\big)}. \tag{2}$$

*Proof.* Lemma S1.1 applies verbatim (its proof used only independence and Gaussianity of the *noise*). With conditionally i.i.d. snapshots, $I(\mathcal{S}; \boldsymbol{C}^N) \leq N\, I(\mathcal{S}; \boldsymbol{c}_1)$, and with conditionally independent coordinates, subadditivity of entropy gives $I(\mathcal{S}; \boldsymbol{c}) \leq \sum_\tau \big[h(c_\tau) - h(c_\tau \mid \mathcal{S})\big]$. Under the uniform prior each coordinate is active with probability $k/p$, so $\mathrm{Var}(c_\tau) = v_0(1 + \rho k/p)$ and the Gaussian maximum-entropy bound gives $h(c_\tau) \leq \frac{1}{2}\log(2\pi e\, v_0(1 + \rho k/p))$. Conditionally on $\mathcal{S}$, $c_\tau$ is an independent signal plus $\mathcal{N}(0, v_0)$ noise, so $h(c_\tau \mid \mathcal{S}) \geq \frac{1}{2}\log(2\pi e\, v_0)$ (adding an independent variable cannot decrease entropy). Summing over the $p$ coordinates yields $I(\mathcal{S}; \boldsymbol{c}) \leq \frac{p}{2}\log(1 + \rho k/p)$ and Fano concludes. □

**Remark S1.4** (How the two converses compare)**.** *Both bounds are valid for Gaussian signals, so one may take their pointwise maximum. For sparse supports ($k \ll p$) and small $\rho$ the Gaussian bound is stronger by a factor $\asymp 1/\rho$ (its denominator is $k\mathrm{KL}_1 \asymp k\rho^2/4$ versus $\asymp k\rho/2$). In the dense regime ($k \asymp p$) the agnostic budget grows only logarithmically in $\rho$, so it* overtakes *the Gaussian-KL bound at high SNR (Fig. 1B). Figure 1A shows the $\log p$ effect: at the scale of the paper's Fig. 1 ($p = 8$) the joint-recovery floor sits well below the per-triangle Chernoff floor, while at $p = 10^4$ it narrows the gap to a $\approx 2\times$ factor (the curves are not directly comparable in level: Fano is normalized at error $1/2$, the Chernoff floor at $\delta = 0.05$).*

**Proposition S1.5** (Sub-Gaussian achievability)**.** *Suppose the curl-coordinate noise and signals are sub-Gaussian with $\psi_2$-norms bounded by $K$ times their standard deviations. Then the energy detector with threshold $\gamma = v_0(1 + \rho/2)$ recovers $\mathcal{S}$ exactly with probability $\geq 1 - \delta$ once*

$$N \;\geq\; \frac{C\,K^4}{\min(\rho, \rho^2)/(1+\rho)^2}\,\log\frac{2p}{\delta},$$

*for a universal constant $C$: the same $\log p/\rho^2$ scaling as the Gaussian theory, with constants degraded by $K^4$.*

*Proof sketch.* Each $c_{\tau,t}^2 - \mathbb{E}c_{\tau,t}^2$ is sub-exponential with $\psi_1$-norm $\lesssim K^2 v_i$. Bernstein's inequality [1, Thm. 2.8.1] gives, for the empirical energy $\hat{E}_\tau = \frac{1}{N}\sum_t c_{\tau,t}^2$,

$$\Pr\big[|\hat{E}_\tau - v_i| \geq t\big] \;\leq\; 2\exp\Big(-cN\min\big(t^2/(K^4 v_i^2),\; t/(K^2 v_i)\big)\Big).$$

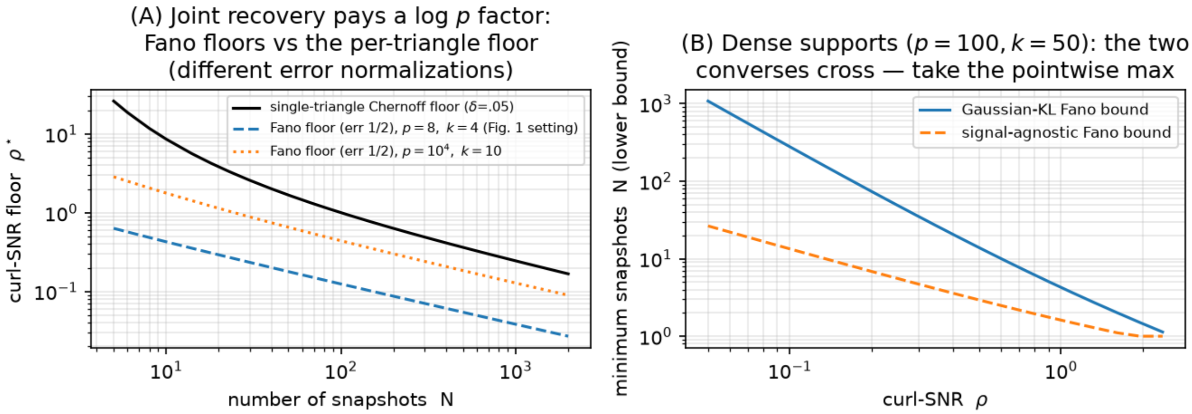


Figure 1: (A) Fano floors (Thm. S1.2, inverted in $\rho$ at error 1/2) against the single-triangle Chernoff floor of the main paper. (B) Dense-support regime: the Gaussian-KL and signal-agnostic bounds cross. Script: `experiments/run_fano.py`.

The two hypotheses are separated by $v_1 - v_0 = v_0\rho$; taking $t = v_0\rho/2$ and a union bound over the $p$ candidates yields the claim. $\square$

## S2 Partial edge observation

Suppose only a subset $\mathcal{E}_{\text{obs}} \subseteq \mathcal{E}$ of edges is observed — each edge independently with probability $q$ (Bernoulli sampling). Write $\boldsymbol{f}_t^{\text{obs}}$ for the observed sub-vector.

**Proposition S2.1** (Observability for curl statistics)**.** *The curl statistic $c_\tau = \boldsymbol{B}_2[:,\tau]^\top \boldsymbol{f}$ of candidate $\tau$ is computable from $\boldsymbol{f}^{\text{obs}}$ iff* all three *edges of $\tau$ are observed. Under Bernoulli($q$) sampling each triangle is observable with probability $q^3$, independently across edge-disjoint candidates.*

An unobservable *active* triangle is an automatic miss; an unobservable *inactive* one is an automatic correct rejection (the detector defaults to "inactive"). Combining with the exact per-triangle marginal laws (main paper, Thm. 2) gives a closed form on edge-disjoint geometries:

**Corollary S2.2** (Exact recovery under edge sampling)**.** *With edge-disjoint candidates, the centered whitened detector at budget $N$ achieves*

$$\Pr[\hat{\mathcal{S}} = \mathcal{S}] = \left[q^3\,(1 - P_{\text{miss}})\right]^k \left[1 - q^3 P_{\text{fa}}\right]^{p-k}, \tag{3}$$

*where $P_{\text{miss}}, P_{\text{fa}}$ are the $\chi^2_{N-1}$ error probabilities of the fully observed test.*

*Proof.* Independence of survival (Prop. S2.1) and of the whitened scores ($\boldsymbol{G} = 3\boldsymbol{I}$). An active triangle is recovered iff it survives (prob. $q^3$) *and* is detected ($1 - P_{\text{miss}}$); an inactive one is mislabeled iff it survives *and* false-alarms ($q^3 P_{\text{fa}}$). $\square$

The $q^{3k}$ factor is brutal: partial sampling multiplies the fully observed recovery probability by $q^{3k} = q^{54}$ at $k = 18$ active triangles — a factor $\approx 0.06$ at $q = 0.95$ and $\approx 0.003$ at $q = 0.9$ — and the Monte-Carlo cliff in Fig. 2B follows the closed form within sampling error. The restricted curve (recovery on the observable candidates only) stays at $\approx 1$ for all $q$: the bottleneck is purely coverage, not detection.

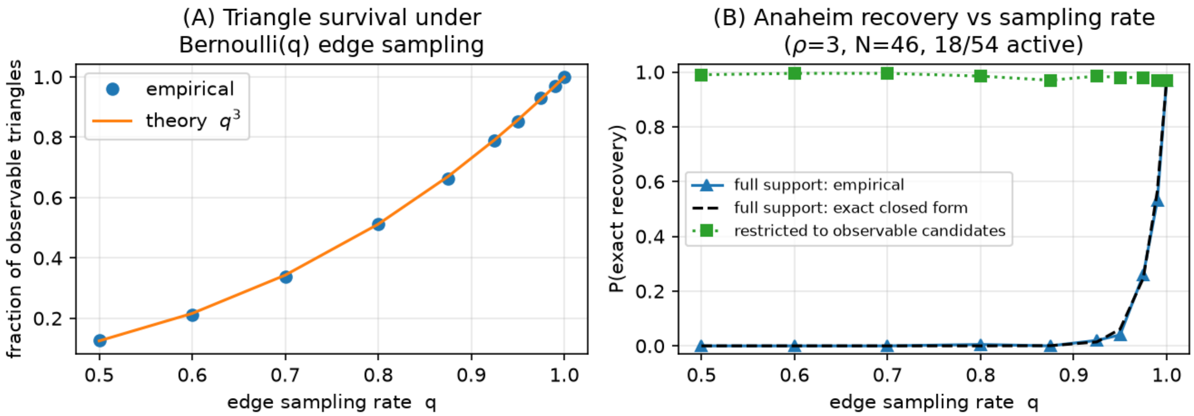


Figure 2: Partial edge observation on Anaheim (real geometry and equilibrium flow background, $\rho = 3$, $N = 46$, 18/54 active). (A) Triangle survival follows $q^3$. (B) Full-support recovery collapses along the exact closed form (3); recovery restricted to observable candidates stays $\approx 1$. Script: `experiments/run_partial_sampling.py`.

**Remark S2.3** (Scope)**.** *Proposition S2.1 is an* operational *statement about detectors built on curl statistics — the class this paper analyzes. A triangle with two observed edges still injects signal into those edges' flows, but that signal is entangled with the unknown gradient nuisance once the exact curl cancellation is unavailable; whether (and at what cost) it can be exploited, e.g. by marginalizing the gradient over the observed sub-graph, is an open question we do not claim to settle. Low-rank flow completion under sampling (cf. graph sampling theory [2]) is the natural attack.*

## S3 Plug-in variance estimation and the adaptive detector

The paper's detectors take $(\sigma_c, \sigma_n)$ as known. We now estimate both from the same data. Let $u_\tau = \mathrm{score}_\tau/(\boldsymbol{G}^{-1})_{\tau\tau}$ denote the normalized whitened scores, so that for inactive $\tau$, $N u_\tau/\sigma_n^2 \sim \chi_d^2$ ($d = N$ or $N-1$ if centered), while active scores are stochastically larger.

**Estimators.**

$$\hat{\sigma}_n^2 = \frac{N\,\mathrm{med}_\tau(u_\tau)}{\mathrm{med}(\chi_d^2)}, \qquad \text{(median, robust to} < 50\% \text{ contamination)} \tag{4}$$

$$\hat{\sigma}_c^2 = \Big(\tfrac{1}{|\hat{\mathcal{A}}|}\textstyle\sum_{\tau\in\hat{\mathcal{A}}}\big[\mathrm{score}_\tau - \hat{v}_{0,\tau}\big]\Big)_+, \qquad \hat{\mathcal{A}} = \text{Bonferroni screen at level } \alpha, \tag{5}$$

followed by one refinement pass: after a first detection, $\sigma_n$ is re-fit by the mean of the *detected* off-support scores and $\sigma_c$ by the on-support excess, and the detector is re-run once. Because the refit set is selected by the same data, the refit is conditionally biased (missed active triangles inflate $\hat{\sigma}_n$ — measured $\approx +5\%$ at $N = 10$ on the strip benchmark, vanishing for $N \geq 20$); its accuracy is established empirically (Fig. 3A), not by an unbiasedness argument, and Prop. S3.2 applies conditionally on the deterministic error bound $\epsilon$.

**Lemma S3.1** (Median consistency envelope)**.** *Assume the candidates are pairwise edge-disjoint, so that the inactive normalized scores are i.i.d.* $\sigma_n^2\chi_d^2/N$ *(the DKW step below requires independence; for correlated scores the envelope is validated only empirically). Let a fraction* $\pi < 1/2$ *of the* $p$ *candidates be active and let* $\epsilon_p = \sqrt{\log(2/\delta)/(2(1-\pi)p)}$ *satisfy* $\pi + \epsilon_p < 1/2$*. Then with probability at least* $1-\delta$*,*

$$q_d\Big(\tfrac{1/2-\pi-\epsilon_p}{1-\pi}\Big) \;\le\; \frac{N\,\mathrm{med}_\tau(u_\tau)}{\sigma_n^2} \;\le\; q_d\Big(\tfrac{1/2}{1-\pi} + \tfrac{\epsilon_p}{1-\pi}\Big),$$

*where* $q_d(\cdot)$ *is the* $\chi_d^2$ *quantile function. Consequently* $\hat\sigma_n^2/\sigma_n^2 \in [q_d(\frac{1/2-\pi-\epsilon_p}{1-\pi}), q_d(\frac{1/2+\epsilon_p}{1-\pi})]/q_d(1/2)$ *— an explicit, numerically computable envelope that shrinks to* $\{1\}$ *as* $p, d \to \infty$ *whenever* $\pi < 1/2$*.*

*Proof.* Active scores stochastically dominate inactive ones, so the sample median of all $p$ scores is sandwiched between order statistics of the $(1-\pi)p$ inactive scores: at least a $(1/2-\pi)/(1-\pi)$-fraction and at most a $(1/2)/(1-\pi)$-fraction of the inactive sample lies below it. The Dvoretzky–Kiefer–Wolfowitz inequality applied to the inactive empirical CDF converts these fractions into population $\chi_d^2$ quantiles up to $\epsilon_p$, with probability $1-\delta$. □

**Proposition S3.2** (Plug-in perturbation)**.** *Let the plug-in variances satisfy* $\hat v_{i,\tau} = (1+e_{i,\tau})v_{i,\tau}$ *with* $|e_{i,\tau}| \le \epsilon < 1/2$*. Then each per-triangle error probability of the adaptive detector exceeds its known-parameter counterpart by at most* $L_d\,\epsilon$*, where* $L_d < \infty$ *is a Lipschitz constant of the* $\chi_d^2$ *CDFs at the optimal threshold, and the exact-recovery probability drops by at most* $p\,L_d\,\epsilon$ *(union bound).*

*Proof sketch.* The Bayes threshold $\gamma(v_0, v_1)$ of the two-variance test is a smooth function of $(v_0, v_1)$ (main paper, Prop. 1), so $|\hat\gamma - \gamma| \lesssim \epsilon\gamma$; the error probabilities are integrals of bounded $\chi_d^2$ densities over the threshold shift. □

Empirically (Fig. 3) both estimates contract at the $1/\sqrt{N}$ rate and the fully adaptive detector's recovery curve is indistinguishable from the known-parameter detector for $N \gtrsim 30$ on the confusable strip benchmark; the residual gap at $N \le 20$ (at most 0.06 after refinement) is the price of estimating a variance from nine medians.

# S4 Reproducibility

All figures regenerate on CPU from fixed seeds via `experiments/run_all.py`; individually: Fig. 1 ← `run_fano.py` (pure theory, seconds); Fig. 2 ← `run_partial_sampling.py` (≈4 min); Fig. 3 ← `run_plugin.py` (≈3 min). The test-suite (`pytest -q`) checks: the Fano bounds against the empirical phase boundary of the main paper's Fig. 1 (a converse must lower bound it), the closed form (3) against Monte-Carlo, the envelope of Lemma S3.1, and the adaptive-vs-known recovery gap.

# References


[1] R. Vershynin, *High-Dimensional Probability*, Cambridge Univ. Press, 2018.

[2] S. Chen, R. Varma, A. Sandryhaila, and J. Kovačević, "Discrete signal processing on graphs: Sampling theory," *IEEE Trans. Signal Process.*, vol. 63, no. 24, pp. 6510–6523, 2015.

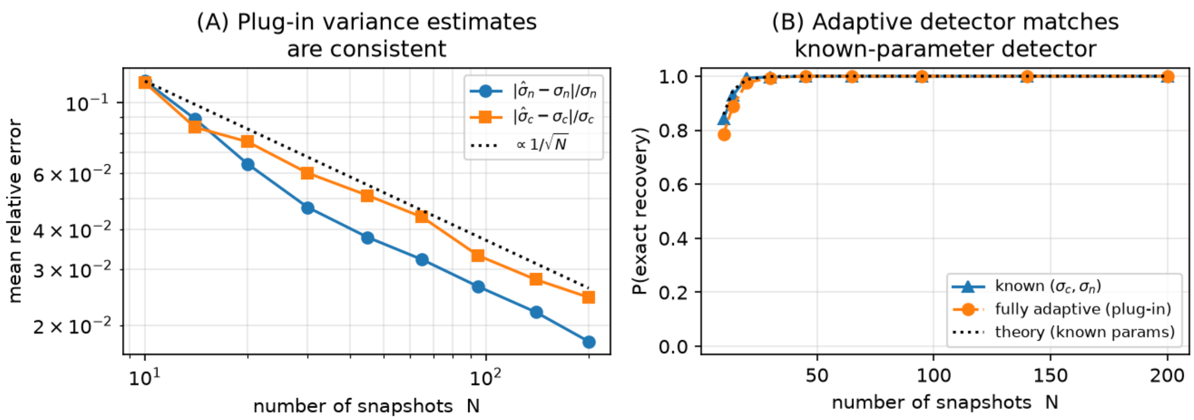


Figure 3: Plug-in estimation on the strip benchmark of the main paper's Fig. 2 ($\rho = 8$, 4/9 active). (A) Consistency of (4)–(5) (after refinement). (B) The fully adaptive detector matches the known-parameter detector. Script: `experiments/run_plugin.py`.